

Money Matrix: Methodology Report

1 Introduction

This report explains the algorithmic trading strategy built for the Money Matrix problem. The main goal of our strategy was to beat the standard "Buy and Hold" benchmark by using a mix of technical indicators.

2 Core Hypothesis

The fundamental hypothesis of our strategy is that stock markets move in trends, but they also face sudden crashes and periods of high volatility. A simple "Buy and Hold" strategy forces you to suffer through massive losses during market crashes.

We hypothesized that by combining three different types of indicators—a trend follower (MACD), a momentum checker (RSI), and a volatility filter (Supertrend)—we can actively decide when it is safe to be in the market. By exiting during bad times and riding the trend during good times, we can achieve significantly better returns with lower risk.

3 Technical Indicators Used

Relying on just one indicator can give false signals. To make our strategy safe, we combined three different indicators:

3.1 1. MACD (Moving Average Convergence Divergence)

What it is: A trend-following indicator showing the relationship between two moving averages of a stock's price.

Why we used it: We used MACD to catch the main trend early. If the MACD line crosses above the signal line, it tells us the trend is going up (Bullish). If it crosses below, the trend is going down (Bearish).

3.2 2. RSI (Relative Strength Index)

What it is: A momentum indicator that measures the speed of price changes on a scale of 0 to 100.

Why we used it: Sometimes a stock's price goes up too fast and it becomes "overbought", meaning it might fall soon. We used RSI to check the strength of the move and avoid buying at the very top.

3.3 3. Supertrend

What it is: A trend-following indicator placed directly on the price chart that uses the Average True Range (ATR) to account for market volatility.

Why we used it: Supertrend acts as a great filter and a dynamic stop-loss. We only take buy trades if the price is above the Supertrend line, protecting us from sudden market crashes.

4 Parameter Choices

When building an algorithmic strategy, it is very easy to cheat by testing hundreds of random numbers until you find the exact combination that gives the highest past profit. This is called "curve-fitting" or over-fitting, and such strategies fail in the real world.

To avoid this, we chose the standard, globally accepted default parameters for all our indicators:

- **MACD:** Fast Length = 12, Slow Length = 26, Signal = 9.
- **RSI:** Time Period = 14 days.
- **Supertrend:** ATR Period = 10, Multiplier = 3

Justification: Using standard parameters proves that the core logic of our voting system is strong on its own, and we are not just manipulating numbers to force a good result on historical data.

5 Signal Generation Logic (The Voting System)

To make our final decision, we created a simple "Voting System". Each of the three indicators gets to cast a vote:

- **+1 Vote (Bullish):** If the indicator shows a buy signal.
- **-1 Vote (Bearish):** If the indicator shows a sell signal.
- **0 Vote (Neutral):** If there is no clear signal.

We sum up the votes to get a Composite Score for each day:

- **BUY Signal:** If the total score is ≥ 2 (Majority says market is going up).
- **SELL Signal:** If the total score is ≤ -2 (Majority says market is going down).
- **HOLD / Clear Position:** If the score is between -1 and 1, we hold cash because the market is confused.

Justification: This voting logic ensures that we only enter a trade when at least two major indicators agree. It heavily reduces fake signals and saves us from unnecessary losses.

6 Backtesting Methodology

To rigorously validate our hypothesis, we implemented a comprehensive backtesting framework prior to signal generation:

- **Data & Universe:** We fetched 7 years of daily adjusted close prices using the `yfinance` library, covering 10 highly liquid stocks across 5 different sectors (Technology, Financials, Industrials, Energy, and Healthcare).
- **Capital & Costs:** The starting capital was assumed to be \$1,000,000. To simulate real-world trading friction, a transaction cost of 10 basis points (0.10%) was applied to every trade.
- **Data Preprocessing:** A conservative 250-trading-day warm-up buffer was utilized to ensure all long-term indicators were accurately initialized before generating any market signals.

7 Tackling Overfitting and Ensuring Robustness

To guarantee that our strategy performs in live markets and isn't just optimized for historical success, we tackled the overfitting problem in three specific parts:

7.1 Part 1: The Risk of Curve-Fitting

In quantitative finance, curve-fitting occurs when a model is excessively complex and tailored to historical noise rather than the actual trend. A single indicator strategy is highly vulnerable to this. If parameters are tweaked just to maximize past returns, the strategy will inevitably collapse when future market dynamics shift.

7.2 Part 2: Prevention via The Voting Mechanism

By integrating three different mathematical dimensions (trend, momentum, and volatility) and utilizing universally accepted default parameters (as highlighted in Section 5), we structurally prevented parameter manipulation. Requiring a minimum composite score of ± 2 ensures the system ignores isolated, noisy market movements and only acts on statistically significant consensus.

7.3 Part 3: Strategy Robustness (Asset & Time)

A strategy is robust if it works consistently across different scenarios, not just on one specific stock. We ensured robustness in two ways:

- **Asset Robustness (Diversification):** We ran the exact same rules on 10 different companies from various sectors. The system adapted well to all of them.
- **Time Robustness:** We backtested the data over a long period of 7 years. This timeframe includes massive bull markets as well as the sudden 2020 COVID-19 crash. Surviving different market conditions proves the reliability of our system.

8 Performance, Results, and Alpha Generated

In algorithmic trading, "Alpha" refers to the extra profit your strategy makes compared to a simple benchmark (like just buying and holding the stocks).

We tested our strategy over the historical data and compared it with an Equal-Weight Buy & Hold benchmark.

- **Initial Capital:** \$1,000,000
- **Benchmark Profit:** \$6,320,128
- **Our Strategy Profit:** \$8,372,504

The difference between these two numbers is **\$2,052,376**. This extra \$2+ million is our Alpha. This large Alpha proves that our logic of timing the market entries and exits successfully added massive financial value.

9 Conclusion

Our strategy successfully outperformed the benchmark by generating over \$2 million in Alpha. By keeping the logic simple and relying on a multi-indicator voting system, we managed to capture big trends while avoiding periods of high volatility. The combination of MACD (for trend), RSI (for momentum), and Supertrend (for volatility filtering), backed by rigorous backtesting, proved to be highly effective, robust, and free from overfitting.